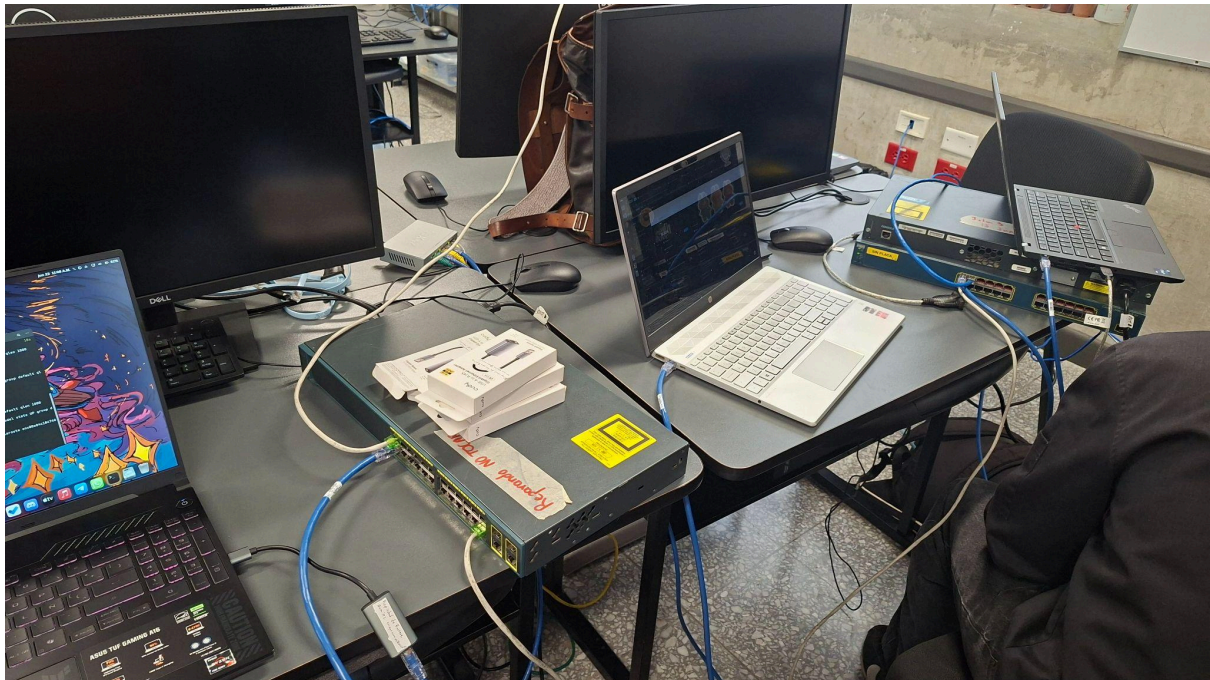


Pruebas de conexión:

Pruebas hechas en equipos físicos:



Intermediario:

```
test - Antigravity IDE - Makefile
Explorer
test
intermediario
bitacora.log
intermediario
intermediario.cpp
intermediario.h
Logger.h
main.cpp
Makefile
README.md
Socket.cc
Socket.h
VSocket.cc
VSocket.h
intermediario Enrique
NachoSx64
servidor
servidor Enrique
arrancar.sh
GUIA_PRUEBAS.md
intermediario.zip
limpiar.sh
Makefile
Conectar Intermediarios ...
intermediario.cpp
intermediario: $(ALL_SRC) intermediario.h Logger.h Socket.h VSocket.h
25
26
27 run: intermediario
28 ./intermediario 0.0.0.0 8081 172.16.123.35 8080 192.168.1.255
29
30 run_local: intermediario
31 ./intermediario 0.0.0.0 8081 auto 8080 255.255.255.255 127.0.0.1 2026
Problems Output Debug Console Terminal Ports
~/Documents/PIRO/test/intermediario
make run
[INTERMEDIARY] [23/06/2026 10:55] [START] bindIp=0.0.0.0 clientPort=8081 figureServer=172.16.123.35:8080
[INTERMEDIARY] [23/06/2026 10:55] [ROUTE_TABLE] construyendo tabla de rutas local desde 172.16.123.35:8080
[INTERMEDIARY] [23/06/2026 10:55] [LISTEN] UDP JOIN en puerto 3030
Intermediario escuchando JOIN UDP en puerto 3030
[INTERMEDIARY] [23/06/2026 10:55] [BROADCAST] Iniciando broadcast JOIN periodico
Intermediario escuchando clientes en 0.0.0.0:8081
[INTERMEDIARY] [23/06/2026 10:55] [LISTEN] UDP descubrimiento de intermediarios remotos en puerto 2027
[INTERMEDIARY] [23/06/2026 10:55] [LISTEN] clientes en 0.0.0.0:8081
[INTERMEDIARY] [23/06/2026 10:55] [PEER ANNOUNCE] Anunciando 'INTER_HERE 8081' a 192.168.1.255:2027
[INTERMEDIARY] [23/06/2026 10:55] [LISTEN] TCP PEER en puerto 3031
Intermediario escuchando peers TCP en puerto 3031
[INTERMEDIARY] [23/06/2026 10:55] [ROUTE_TABLE] Figuras locales registradas: 7 (total en tabla: 7)

===== TABLA DE RUTAS =====
Figura IP Servidor Puerto
-----
death_star_ii 172.16.123.35 8080
[INTERMEDIARY] [23/06/2026 10:55] [ROUTE] figura=death_star_ii ip=172.16.123.35 puerto=8080
dragon 172.16.123.35 8080
[INTERMEDIARY] [23/06/2026 10:55] [ROUTE] figura=dragon ip=172.16.123.35 puerto=8080
fish 172.16.123.35 8080
[INTERMEDIARY] [23/06/2026 10:55] [ROUTE] figura=fish ip=172.16.123.35 puerto=8080
giraffe 172.16.123.35 8080
[INTERMEDIARY] [23/06/2026 10:55] [ROUTE] figura=giraffe ip=172.16.123.35 puerto=8080
horse 172.16.123.35 8080
[INTERMEDIARY] [23/06/2026 10:55] [ROUTE] figura=horse ip=172.16.123.35 puerto=8080
imperial_destroyer 172.16.123.35 8080
[INTERMEDIARY] [23/06/2026 10:55] [ROUTE] figura=imperial_destroyer ip=172.16.123.35 puerto=8080
millennium_falcon 172.16.123.35 8080
[INTERMEDIARY] [23/06/2026 10:55] [ROUTE] figura=millennium_falcon ip=172.16.123.35 puerto=8080
=====
[INTERMEDIARY] [23/06/2026 10:55] [PEER DISCOVERED] 172.16.123.51:2026
[INTERMEDIARY] [23/06/2026 10:55] [HTTP_PEER_ADDED] 172.16.123.51:2026
Ln 28, Col 1 Tab Size: 4 UTF-8 LF Makefile Antigravity - Settings
```

Conexión con servidor, descubrimiento con intermediario de isla 3

```

intermediario > Makefile
16 intermediario: $(ALL_SRC) intermediario.h Logger.h Socket.h VSocket.h
25
26 run: intermediario
27     ./intermediario 0.0.0.0 8081 172.16.123.35 8080 192.168.1.255
28     Ctrl-H for Agent
29 run_local: intermediario
30     ./intermediario 0.0.0.0 8081 auto 8080 255.255.255.255 127.0.0.1 2026
31
32 run_ip:
33     ./intermediario 0.0.0.0 8081 auto 8080 255.255.255.255 10.69.199.11 2026
34

Problems Output Debug Console Terminal Ports
~/Documents/PIRO/test/intermediario
2m 3s

< make run
[INTERMEDIARY] [23/06/2026 10:55] [LISTEN] UDP descubrimiento de intermediarios remotos en puerto 2027
[INTERMEDIARY] [23/06/2026 10:55] [LISTEN] Clientes en 0.0.0.0:8081
[INTERMEDIARY] [23/06/2026 10:55] [PEER_ANNOUNCE] Anunciando 'INTER_HERE 8081' a 192.168.1.255:2027
[INTERMEDIARY] [23/06/2026 10:55] [LISTEN] TCP PEER en puerto 3031
Intermediario escuchando peers TCP en puerto 3031
[INTERMEDIARY] [23/06/2026 10:55] [ROUTE_TABLE] Figuras locales registradas: 7 (total en tabla: 7)

===== TABLA DE RUTAS =====
Figura      IP Servidor  Puerto
-----
death_star_ii 172.16.123.35 8080
[INTERMEDIARY] [23/06/2026 10:55] [ROUTE] figura=death_star_ii ip=172.16.123.35 puerto=8080
dragon        172.16.123.35 8080
[INTERMEDIARY] [23/06/2026 10:55] [ROUTE] figura=dragon ip=172.16.123.35 puerto=8080
fish          172.16.123.35 8080
[INTERMEDIARY] [23/06/2026 10:55] [ROUTE] figura=fish ip=172.16.123.35 puerto=8080
giraffe       172.16.123.35 8080
[INTERMEDIARY] [23/06/2026 10:55] [ROUTE] figura=giraffe ip=172.16.123.35 puerto=8080
horse         172.16.123.35 8080
[INTERMEDIARY] [23/06/2026 10:55] [ROUTE] figura=horse ip=172.16.123.35 puerto=8080
imperial_destroyer 172.16.123.35 8080
[INTERMEDIARY] [23/06/2026 10:55] [ROUTE] figura=imperial_destroyer ip=172.16.123.35 puerto=8080
millennium_falcon 172.16.123.35 8080
[INTERMEDIARY] [23/06/2026 10:55] [ROUTE] figura=millennium_falcon ip=172.16.123.35 puerto=8080
=====

[INTERMEDIARY] [23/06/2026 10:55] [PEER_DISCOVERED] 172.16.123.51:2026
[INTERMEDIARY] [23/06/2026 10:55] [HTTP_PEER_ADDED] 172.16.123.51:2026
[INTERMEDIARY] [23/06/2026 10:55] [HTTP_PEER_MERGED] 172.16.123.51:2026 figuras_nuevas=3
[INTERMEDIARY] [23/06/2026 10:55] [CLIENT_REQUEST] GET /lego/list.php?figure=horse&part=2 HTTP/1.1
[INTERMEDIARY] [23/06/2026 10:55] [HANDLE_LIST] figure=horse part=2
[INTERMEDIARY] [23/06/2026 10:55] [PROTO_REQUEST] Enviando: P/G/horse:2 -> 172.16.123.35:8080

```

Solicitud de figuras desde cliente y otros intermediarios

```

~/Documents/PIRO/test/intermediario
2m 3s

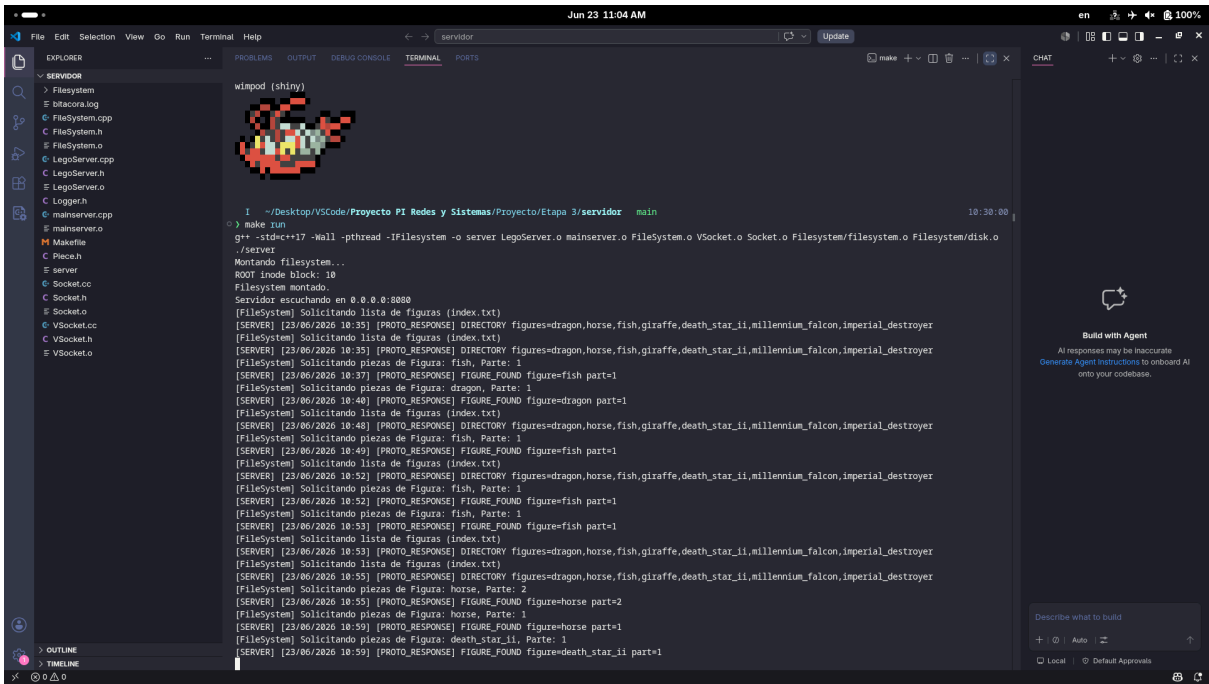
< make run
[INTERMEDIARY] [23/06/2026 10:55] [ROUTE] figura=dragon ip=172.16.123.35 puerto=8080
fish          172.16.123.35 8080
[INTERMEDIARY] [23/06/2026 10:55] [ROUTE] figura=fish ip=172.16.123.35 puerto=8080
giraffe       172.16.123.35 8080
[INTERMEDIARY] [23/06/2026 10:55] [ROUTE] figura=giraffe ip=172.16.123.35 puerto=8080
horse         172.16.123.35 8080
[INTERMEDIARY] [23/06/2026 10:55] [ROUTE] figura=horse ip=172.16.123.35 puerto=8080
imperial_destroyer 172.16.123.35 8080
[INTERMEDIARY] [23/06/2026 10:55] [ROUTE] figura=imperial_destroyer ip=172.16.123.35 puerto=8080
millennium_falcon 172.16.123.35 8080
[INTERMEDIARY] [23/06/2026 10:55] [ROUTE] figura=millennium_falcon ip=172.16.123.35 puerto=8080
=====

[INTERMEDIARY] [23/06/2026 10:55] [PEER_DISCOVERED] 172.16.123.51:2026
[INTERMEDIARY] [23/06/2026 10:55] [HTTP_PEER_ADDED] 172.16.123.51:2026
[INTERMEDIARY] [23/06/2026 10:55] [HTTP_PEER_MERGED] 172.16.123.51:2026 figuras_nuevas=3
[INTERMEDIARY] [23/06/2026 10:55] [CLIENT_REQUEST] GET /lego/list.php?figure=horse&part=2 HTTP/1.1
[INTERMEDIARY] [23/06/2026 10:55] [HANDLE_LIST] figure=horse part=2
[INTERMEDIARY] [23/06/2026 10:55] [PROTO_REQUEST] Enviando: P/G/horse:2 -> 172.16.123.35:8080
[INTERMEDIARY] [23/06/2026 10:55] [FIGURE_FOUND] figure=horse part=2
[INTERMEDIARY] [23/06/2026 10:55] [PEER_DISCOVERED] 172.16.123.51:2026
[INTERMEDIARY] [23/06/2026 10:55] [PEER_DISCOVERED] 172.16.123.51:2026
[INTERMEDIARY] [23/06/2026 10:55] [PEER_DISCOVERED] 172.16.123.51:2026
[INTERMEDIARY] [23/06/2026 10:55] [CLIENT_REQUEST] GET /lego/index.php HTTP/1.1
[INTERMEDIARY] [23/06/2026 10:55] [HANDLE_INDEX] Figuras disponibles: 10
[INTERMEDIARY] [23/06/2026 10:55] [PEER_DISCOVERED] 172.16.123.51:2026
[INTERMEDIARY] [23/06/2026 10:55] [PEER_DISCOVERED] 172.16.123.51:2026
[INTERMEDIARY] [23/06/2026 10:55] [CLIENT_REQUEST] GET /lego/list.php?figure=furro&part=1 HTTP/1.1
[INTERMEDIARY] [23/06/2026 10:55] [HANDLE_LIST] figure=furro part=1
[INTERMEDIARY] [23/06/2026 10:55] [REMOTE_REQUEST] GET /figura/furro/1 -> 172.16.123.51:2026
[INTERMEDIARY] [23/06/2026 10:55] [FIGURE_FOUND] figure=furro part=1
[INTERMEDIARY] [23/06/2026 10:55] [PEER_DISCOVERED] 172.16.123.51:2026

```

Tabla de rutas creada exitosamente

Servidor



Cliente

